

ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA
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Sommario

TODO IoT (Internet of Things) è l'acronimo utilizzato per indicare la sfera tecnologica che ha alla base l'utilizzo di oggetti intelligenti. Per oggetti intelligenti si intendono quei dispositivi, macchine o attrezzature che si possono connettere a internet per trasmettere, ricevere dati o elaborarli. All'interno di questa sfera, uno dei metodi di comunicazione più utilizzati è quello basato su un modello Publish/Subscribe che consente lo scambio di dati tra dispositivi della rete senza la necessità di connessioni dirette tra loro. Per funzionare si utilizza il protocollo MQTT, uno dei protocolli più diffusi per lo scambio di messaggi, il quale si basa su una struttura con 3 tipologie di dispositivi connessi: Publisher, Subscriber e Broker. I Publisher una volta ottenuti dei dati tramite sensori di vario genere, li pubblica, rendendoli disponibili nella rete. I dati per spostarsi all'interno della rete utilizzano i broker cioè vari punti/server di connessione, i quali permettono il passaggio di dati per poi infine arrivare ai Subscribers per la lettura e l'elaborazione. Questa tesi riprende il lavoro effettuato da un collega nell'anno 21-22 approfondendo e sviluppando i casi di test con particolare attenzione a quelle che sono le caratteristiche che possono portare criticità al calcolo della soluzione ottima di percorso per il flusso dei dati. La tesi del collega si basa sull'idea di costruire un sistema distribuito sfruttando le proprietà matematiche del rilassamento Lagrangiano, dove ogni nodo conosce solamente i propri vicini.

Capitolo 1

Introduzione

1.1 Iot (Internet of Things)

Il termine Iot nasce come acronimo per definire quella sfera tecnologica moderna che si basa sull'utilizzo di dispositivi "intelligenti". Un dispositivo definito intelligente o più sentito comunemente "smart" è un dispositivo in grado di connettersi alla rete per poter svolgere diverse funzioni, tra cui la più importante è lo scambio di informazioni con altri dispositivi. Nella mia tesi mi soffermo particolarmente sui "sensori", dispositivi chiave per lo scambio di messaggi in una rete Iot. Un sensore è un dispositivo che registra un cambiamento nel mondo fisico, ne genera un messaggio virtuale e poi diffonde il messaggio oppure prende decisioni tramite il valore letto. Nella mia tesi vado ad approfondire proprio questo dettaglio. Il fatto che un sensore possa generare un messaggio e permettere ad altri dispositivi di prendere decisioni tramite esso significa che se le decisioni devono essere tempestive allora anche il metodo di trasmissione e l'efficienza della stessa lo devono essere.

1.2 Publish and Subscribe

Tra le metodologie più famose di trasmissione di messaggi quella su cui mi voglio soffermare è quella del Publish and Subscribe, un paradigma particolarmente adatto all'Iot, nato come sostituzione del classico Client-Server. In questo schema i dispositivi si dividono in Publisher e Subscriber collegati tra loro tramite dei nodi/server di tra-

1.2. PUBLISH AND SUBSCRIBE

smissione definiti Broker. In questo modello ognuno a il suo ruolo: I Publisher devono diffondere nella rete le informazioni in loro possesso, i Broker devono ritrasmettere queste informazioni senza permettere danneggiamenti o perdita delle stesse e i Subscriber sono i ricevitori o i richiedenti delle tali. Il meccanismo alla base è molto semplice: Il Publisher invia messaggi etichettati con un topic, il broker li riceve e li inoltra solamente ai Subscriber iscritti a quel topic. Questo meccanismo permette un disaccoppiamento tra nodo di partenza e di arrivo in quanto le 2 parti non si conoscono direttamente e non devono per forza essere attivi contemporaneamente. Il protocollo standard per la implementazione di questo modello è l'MQTT, progettato per essere leggero e robusto anche su reti instabili.

Structure of the Thesis

Capitolo 2

State of the art

I suggest referencing stuff as follows: fig. 2.1 or Figure 2.1

2.1 Some cool topic

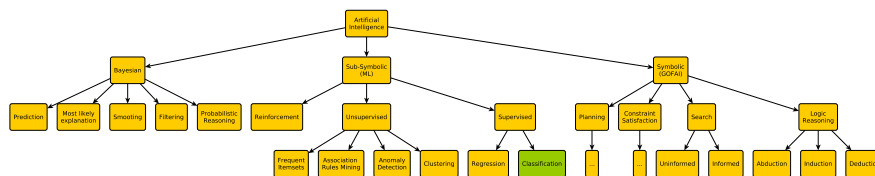


Figura 2.1: Some random image

Capitolo 3

Contribution

You may also put some code snippet (which is NOT float by default), eg: chapter 3.

3.1 Fancy formulas here

```
1 public class HelloWorld {  
2     public static void main(String[] args) {  
3         // Prints "Hello, World" to the terminal window.  
4         System.out.println("Hello, World");  
5     }  
6 }
```

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